Workforce Tracking by Role, Operation and Snapshot

Oct 21, 2026 · @LMS

Purpose and how to read this

This document shows who is working on which operation, who has not started, who has finished, and who is responsible for each activity, at every 5-minute snapshot of the shift day **15 October 2026** (06:00 to 23:30, 211 snapshots).

All names, roles, assignments and times come from the GiNi workbooks (worker master, plant worker snapshots, operation instances, crew intervals and task assignments). **No invented sample data was needed**; the few values that are derived rather than stored are marked in the Gaps section.

- **196 people** are in the worker master: 160 production operators, 12 quality inspectors, 8 maintenance technicians, 8 material handlers, and 8 supervisors and managers.
- The **188 shop-floor workers** (the first four groups) are given a status at every snapshot. Supervisors and managers are shown as the people *responsible* for the work, not given a status.
- Two shifts: **Shift A** 06:00 to 14:30 and **Shift B** 14:30 to 23:00, each with 94 shop-floor workers.
- Four work centres do the labour: Pre-Assembly (WC1), Assembly (WC2), Painting (WC3) and Markings (WC4). Operation OP10 to OP40 runs in that order, with inspection gates QG20, QG30 and QG40 between them.

Worker roles

Group	Role	Total	Shift A	Shift B	Where they work
Operators	Production operator	160	80	80	40 in each of WC1 Pre-Assembly, WC2 Assembly, WC3 Painting, WC4 Markings
Inspectors	Quality inspector	12	6	6	Plant-wide, at the QG20, QG30 and QG40 gates

Group	Role	Total	Shift A	Shift B	Where they work
Other roles	Maintenance technician	8	4	4	Plant-wide, on call to any work centre
Other roles	Material handler	8	4	4	Plant-wide, indirect support
Responsible	Area supervisor	4	2	2	Two work centres each
Responsible	Shift manager	2	1	1	All four work centres
Responsible	Quality supervisor	2	1	1	All four work centres

Operators take a trade on each task: Fitter, Welder, Assembler, Painter, Marker, or Trainee. A person can hold a different trade on a different task, so the trade is shown with the assignment and not as a fixed role.

Who is responsible for each activity

Shift	Person	Role	Responsible for
A	Joshua Watson (W0175)	Area supervisor	WC1 Pre-Assembly and WC2 Assembly: assigns every operator task there
A	Eric Wallace (W0177)	Area supervisor	WC3 Painting and WC4 Markings: assigns every operator task there
B	Heather Powell (W0176)	Area supervisor	WC1 and WC2 on the evening shift
B	Samantha Tucker (W0178)	Area supervisor	WC3 and WC4 on the evening shift
A / B	James Nelson (W0173) / Jessica Murphy (W0174)	Shift manager	The whole plant for their shift
A / B	Zachary Anderson (W0179) / Allison Hall (W0180)	Quality supervisor	The inspectors and the quality gates for their shift

In the task records every one of the 11,730 operator tasks was assigned by an area supervisor, which is how the "responsible" column in the later tables is filled.

How status is decided

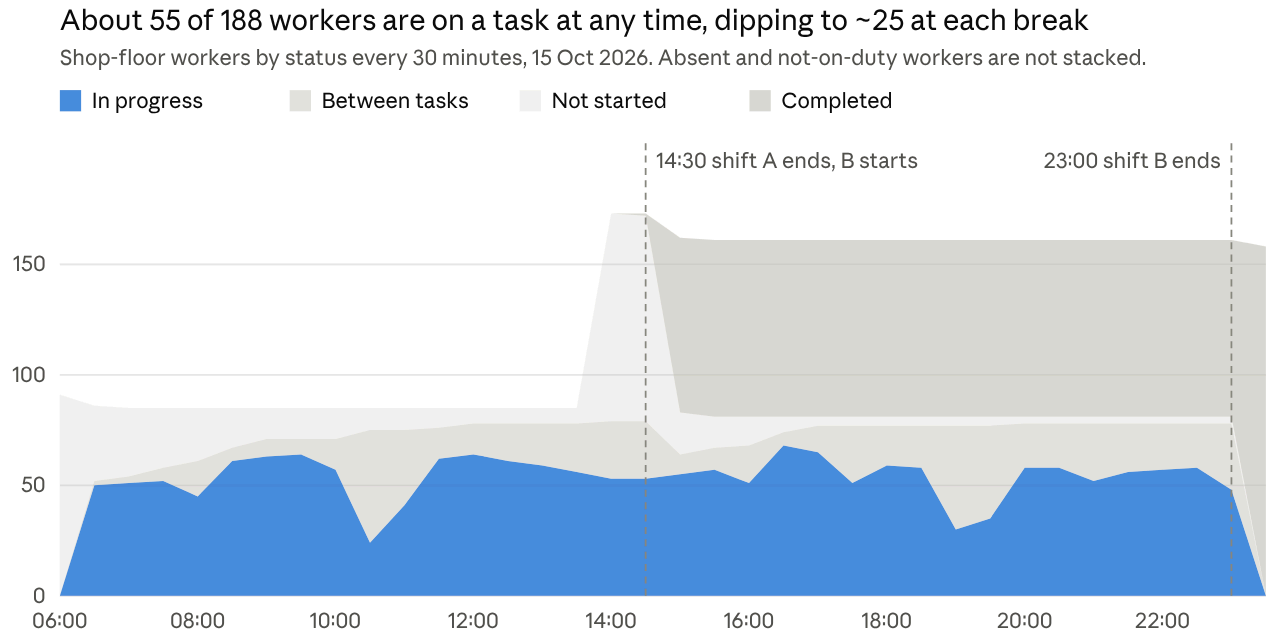
Each shop-floor worker gets one of six statuses at every snapshot. The first three are the ones asked for; the other three keep the picture honest for people who are on a break, on leave or absent.

Status	Rule	Typical state in the data
Not Started	On shift or due on shift, and has not yet done any direct task today	Waiting for a unit, orientation, tool preparation, team huddle, before shift start
In Progress	Doing a task right now	Working, reworking or inspecting an operation instance; material handlers on indirect duty
Between tasks	Has worked today, is on shift, and is not on a task at this moment	Planned break, waiting for crew, waiting for an upstream unit, housekeeping
Completed	Has worked today and the shift has ended (A at 14:30, B at 23:00)	Shift ended, clocked out
Not on duty	Not working today, or clocked out without a task	Leave, vacation, day off, offsite training, other shift not yet started
Absent	Confirmed absent for the shift	Sick call, no-show confirmed by phone

Two guards keep statuses consistent with the operation timelines:

- A worker only becomes **Completed** after their shift end. Someone who clocks out in the middle of the shift and returns stays **Between tasks**, and this happens for 2 workers.
- A worker never moves backwards: nobody goes from Completed to working again, from Absent to working, or from Between tasks back to Not Started. All 211 snapshots were checked for this and no violations were found.

Workforce status across snapshots



GiNi workbook WORKER_SNAPSHOT_PLANT, 36 of 211 snapshots (every 30 minutes), 15 Oct 2026

Three things stand out. First, Shift A ramps up fast: 91 workers are Not Started at 06:00, and only 14 are still waiting by 10:00, 10 of them operators in Markings (WC4), which cannot start until painted tanks arrive. Second, at 14:30 the whole of Shift A moves to Completed within 15 minutes while Shift B takes over, so In Progress stays level across the handover. Third, the dips to about 25 (10:30 and 19:00) are planned break windows, when people shift to Between tasks and then return.

Status by role and work centre

The same six statuses at three snapshots show the day's arc: before work starts (06:00), mid-morning on Shift A (10:00), and early evening with Shift B in full swing (18:00). Counts are people.

Snapshot	Group	Not started	In progress	Between tasks	Completed	Not on duty	Absent
06:00	Operators (160)	77	0	0	0	83	0
06:00	Inspectors (12)	6	0	0	0	6	0
06:00	Maintenance (8)	4	0	0	0	4	0

Snapshot	Group	Not started	In progress	Between tasks	Completed	Not on duty	Absent
06:00	Material handlers (8)	4	0	0	0	4	0
10:00	Operators (160)	11	49	11	0	83	6
10:00	Inspectors (12)	0	3	3	0	6	0
10:00	Maintenance (8)	3	1	0	0	4	0
10:00	Material handlers (8)	0	4	0	0	4	0
18:00	Operators (160)	1	54	14	68	11	12
18:00	Inspectors (12)	0	1	3	6	1	1
18:00	Maintenance (8)	3	0	1	2	2	0
18:00	Material handlers (8)	0	4	0	4	0	0

At 10:00, 94 of the 97 people shown as Not on duty are Shift B, which has not arrived; the other 3 are Shift A workers on leave or offsite training. At 18:00 the Completed group is Shift A (80 people) and the In Progress group is Shift B.

Operators by home work centre at 10:00

Work centre	Not started	In progress	Between tasks	Not on duty (mostly Shift B)	Absent
WC1 Pre-Assembly	0	14	3	20	3
WC2 Assembly	0	13	4	22	1
WC3 Painting	1	16	1	21	1
WC4 Markings	10	6	3	20	1

WC4 Markings is the one work centre that is mostly idle in the morning. Its 10 Not Started operators are waiting on a unit (reason code NO_APPROVED_UNIT_UPSTREAM) or doing tool preparation, housekeeping and training until the first painted tank clears its coating inspection. This is the realistic flow of a four-stage line, not a data fault.

Six operators are Absent on Shift A: five sick calls and one no-show confirmed by phone. Seven more are absent on Shift B.

Who is on which operation at 10:00

This is the snapshot at 10:00 on 15 October 2026, during Shift A. Each row is one tank being worked on, with its crew and the supervisor who assigned the work. OP10 is Shell fit-up and pre-assembly, OP20 is Assembly, reinforcement and welding, OP30 is Surface preparation and paint application, and OP40 is Markings, final dressing and documentation.

Operators In Progress (49 people on 14 tanks)

Work centre	Operation	Tank	Workstation	Crew (trade)	Responsible
WC1	OP10	TNK-000597	WS-WC1-03	James Miller W0001, Sarah Thompson W0002, Cody Young W0011, Amanda Moore W0018 (all Fitters)	Joshua Watson W0175
WC1	OP10	TNK-000598	WS-WC1-04	Nicole Campbell W0004 (Trainee); Tiffany Clark W0010, Caitlin Evans W0012, Chad Bailey W0013 (Fitters)	Joshua Watson W0175
WC1	OP10	TNK-000599	WS-WC1-05	Daniel Lewis W0003 (Trainee); Justin Rogers W0005, Brittany Wood W0006, Adam Perry W0007, Vanessa Hughes W0014 (Fitters)	Joshua Watson W0175
WC1	OP10	TNK-000600	WS-WC1-01	Mason Coleman W0015 (Fitter), crew still forming	Joshua Watson W0175
WC2	OP20	TNK-000592	WS-WC2-05	David Myers W0045, Megan Barnes W0046, Brandon Harrison W0047 (Welders); Ethan Hall W0057 (Assembler)	Joshua Watson W0175
WC2	OP20	TNK-000593	WS-WC2-06	Rebecca Johnson W0048, Brian Harris W0049, Elizabeth Nelson W0050, Benjamin Murphy W0051 (all Welders)	Joshua Watson W0175
WC2	OP20	TNK-000594	WS-WC2-04	Brooke Scott W0042, Whitney Anderson W0056 (Welders); Dylan Collins W0043 (Trainee); Michael Cooper W0059, Lauren Bennett W0060 (Assemblers)	Joshua Watson W0175
WC3	OP30	TNK-000575	WS-WC3-04	Nathan Marshall W0093, Erin Carter W0094, Scott Jackson W0095 (Painters)	Eric Wallace W0177

Work centre	Operation	Tank	Workstation	Crew (trade)	Responsible
WC3	OP30	TNK-000576	WS-WC3-05	Kayla Baker W0096, Derek Stewart W0097, Jamie Ward W0098 (Painters)	Eric Wallace W0177
WC3	OP30	TNK-000586	WS-WC3-06	Grant Smith W0087, Ashley King W0088, Kristen Graham W0100 (Painters)	Eric Wallace W0177
WC3	OP30	TNK-000587	WS-WC3-03	Molly Hicks W0086, Andrew Turner W0089, Stephanie Reed W0090 (Painters)	Eric Wallace W0177
WC3	OP30	TNK-000582	WS-WC3-01	Gregory Mitchell W0081, Jordan West W0085, Jason Price W0091 (Painters); Jared James W0083 (Trainee)	Eric Wallace W0177
WC4	OP40	TNK-000577	WS-WC4-02	Christopher Griffin W0131, Courtney Bailey W0136, Travis Ford W0139 (Markers)	Eric Wallace W0177
WC4	OP40	TNK-000578	WS-WC4-03	Dustin Thompson W0125, Garrett Campbell W0127, Jennifer Coleman W0138 (Markers)	Eric Wallace W0177

Inspectors, maintenance and material handlers In Progress (8 people)

Person	Role	Doing	Where	Responsible
Ryan Bell W0161	Quality inspector	Inspecting at the QG20 assembly gate	TNK-000589, WS-WC2-02	Zachary Anderson W0179
Hannah Collins W0166	Quality inspector	Inspecting at the QG30 coating gate	TNK-000579	Zachary Anderson W0179
Melissa Hayes W0162	Quality inspector	Inspecting at the QG40 markings gate	TNK-000580, WS-WC4-01	Zachary Anderson W0179

Person	Role	Doing	Where	Responsible
Gregory Roberts W0181	Maintenance technician	On a maintenance task at the Painting station	WS-WC3-02	James Nelson W0173
Andrew Parker W0189, Stephanie Kelly W0190, Jason Sanders W0191, Danielle Henderson W0192	Material handlers	Indirect material supply	Plant-wide	James Nelson W0173

Everyone else on Shift A at 10:00

- **Not Started (14):** Lindsey Russell W0084 (orientation training), Aaron Ward W0121 (cross-training shadow), Victoria Lewis W0126, Tyler Clark W0133 (tool and fixture preparation), Paige Rogers W0128 (team huddle), Natalie Miller W0140 (housekeeping), and Kelly Ross W0122, Sean Graham W0123, Caleb Wood W0129, Madison Perry W0130, Kimberly Young W0134 (waiting for the first approved unit from upstream); plus maintenance technicians Alyssa Cooper W0182, Jared Bennett W0183 and Lindsey Sullivan W0184, who are on duty and waiting for a call.
- **Between tasks (14):** operators Timothy Hunter W0009, Kevin Green W0019, Amy Edwards W0020 and Danielle Butler W0092 (waiting for crew); Laura Watson W0052, Jeffrey Powell W0053, Andrea Wallace W0054, Bradley Tucker W0055, Rachel Hunter W0132, Kyle Evans W0135 and Patrick Hughes W0137 (no approved unit upstream); and inspectors Jonathan Boyd W0163, Katherine Jones W0164 and Jacob Scott W0165 (awaiting an inspection request).
- **Absent (6):** Christina Griffin W0008, Emily Ford W0016, Matthew Miller W0017, Corey Ross W0099 and Michelle Ellis W0124 (sick call); Abigail Roberts W0058 (no-show confirmed by phone).
- **Not on duty (3):** Shift A workers on planned leave or offsite training.

Individual timelines across snapshots

Six workers show how one person's status moves through the day. Each line is read from the hourly snapshots; a new line starts only when the status, reason or tank changes.

Worker	Timeline
James Miller W0001, operator (Fitter), WC1, Shift A	06:00 Not Started. 07:00 In Progress on OP10, tank TNK-000590. He then moves to a new tank each hour: TNK-000593 at 08:00, TNK-000597 at 09:00 and 10:00, TNK-000601 at 11:00, TNK-000605 at 14:00. 15:00 Completed, clocked out after shift end.
Kelly Ross W0122, operator, WC4 Markings, Shift A	06:00 to 10:00 Not Started: waiting for an approved unit from upstream, tool and fixture preparation, team huddle. 11:00 Between tasks (planned break). 12:00 In Progress on OP40, tank TNK-000579. 13:00 Between tasks (tool preparation). 14:00 In Progress on OP40, tank TNK-000587. 15:00 Completed, shift ended.
Ryan Bell W0161, quality inspector, Shift A	06:00 Not Started. 07:00 Between tasks, awaiting an inspection request. 09:00 In Progress at the QG20 gate on TNK-000582, then TNK-000589 at 10:00. 11:00 In Progress at QG30 on TNK-000584. 12:00 Between tasks. 15:00 Completed.
Christina Griffin W0008, operator, WC1, Shift A	06:00 Not Started, attendance not yet confirmed. 07:00 Absent (sick call) and stays Absent for the rest of the day.
Andrew Parker W0189, material handler, Shift A	06:00 Not Started. 07:00 In Progress on indirect material handling, then support duty from 11:00. 15:00 Completed.
Brittany James W0106, operator, WC3 Painting, Shift B	06:00 Not on duty (other shift). 14:00 Not Started, before shift start. 15:00 In Progress on OP30, tank TNK-000598. 17:00 Between tasks after a mid-shift exit. 18:00 Between tasks in cross-training. 20:00 waiting for an approved unit. 21:00 In Progress on OP30, tank TNK-000604. 23:00 Between tasks (equipment down), then Completed at the 23:15 snapshot.

The Shift B operator is a good test of the status rules: she leaves the line mid-shift, is not marked Completed, and only reaches Completed after Shift B ends.

Consistency checks against operation timelines

Statuses were checked against the operation, crew and task records for the whole day, not only the snapshots shown above.

Check	Result
Every In Progress operator snapshot (9,711 worker-snapshots, 137 operators) names a tank and operation	9,711 of 9,711
That operator was on the operation's crew list at that moment	9,700 of 9,711 (99.9%)
The operation was running at that moment (started, not yet finished)	9,657 of 9,711 (99.4%)
Anyone double-booked on two operations at once	None
Crew joined before the operation was created, or left after it finished	None
Task started before it was assigned, or completed before it started (11,730 tasks)	None
A worker who moves backwards: Completed to working, Absent to present, Between tasks to Not Started	None across 211 snapshots
Every task assigned by an area supervisor on the worker's own shift	Yes: W0175 and W0177 on Shift A, W0176 and W0178 on Shift B (11,730 of 11,730 tasks)

The small gaps in the second and third checks (11 and 54 worker-snapshots) are timing, not contradictions. The snapshot is taken every 5 minutes, so a worker can be shown on an operation for one snapshot after the crew record shows them leaving or the operation finishing, usually by a few minutes. Where a crew record had a gap, the next join was 30 to 43 minutes later, which matches the planned break.

Gaps and derived values

No sample data was invented. These values are derived or assumed, and you should know about them before presenting this to a client:

- **Status names** (Not Started, In Progress, Between tasks, Completed, Not on duty, Absent) are derived from the stored plant state, activity state and shift times using the rules above. The workbook stores states such as WORKING, WAITING, INDIRECT and CLOCKED_OUT, not these labels.
- **Responsibility for inspectors, maintenance and material handlers** is taken from role: quality supervisor for inspectors and shift manager for maintenance and handlers. The task records only name a responsible person (the area supervisor) for operator tasks.
- **Supervisors and managers** have no working status in the data, so they appear as the responsible person and not in the status counts.
- **Absent and Not on duty** are kept out of the chart stack so the three requested statuses are easy to read. They are in the tables.
- **Eight workers** were on duty and clocked out having never been given a direct task (waiting operators and on-call maintenance). They show as Not on duty after shift end, because nothing marks them as Completed.
- **One day only.** The workbooks hold the snapshots for 15 October 2026, so this document shows that day. The live app shows the same view for whichever session is loaded.